A pedagogical description of pseudo spectral DNS of 2D turbulence

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This work presents a reproducible pseudospectral framework for direct numerical simulation of forced two-dimensional turbulence. A fully dealiased Fourier discretization with integrating-factor RK2 time stepping generates statistically stationary states under random band-limited forcing. The formulation, forcing strategy, nonlinear evaluation, and diagnostics—including energy, enstrophy, palinstrophy, and Reynolds numbers—are documented in detail to provide a transparent reference implementation.

I. INTRODUCTION

Turbulence in two spatial dimensions occupies a central place in fluid dynamics, both as a mathematical model with distinctive physical properties and as an effective description of a wide range of quasi–two–dimensional flows found in nature and engineering systems. Since the pioneering work of Kraichnan [1] and Batchelor [2], it has been understood that the dynamics of 2D turbulence differ fundamentally from their three–dimensional counterpart. The simultaneous conservation of energy and enstrophy in the inviscid limit produces a dual–cascade phenomenology: energy is transferred upscale, while enstrophy cascades to progressively smaller scales. This behaviour has been repeatedly confirmed in laboratory experiments, observational studies, and numerical simulations [3–7].

Beyond its theoretical significance, two–dimensional turbulence is directly relevant to a broad range of physical systems. Quasi-2D flows arise in large-scale geophysical dynamics governed by strong stratification or rotation, such as the atmosphere, oceans, and planetary interiors [3, 8]. Thin soap–film experiments replicate nearly inviscid 2D turbulence in controlled laboratory settings [9]. Electromagnetically driven flows in shallow layers [10] and magnetized plasmas [11] provide additional platforms where 2D turbulent dynamics dominate. In each of these systems, the formation of long-lived coherent vortices, intermittent bursts of enstrophy production, and the organisation of the flow around strain-dominated and vorticity-dominated regions are essential phenomena that influence transport, mixing, and large-scale structure [12–15].

Despite the maturity of the theoretical framework, the practical steps required to perform a reproducible, high-quality direct numerical simulation (DNS) of the two-dimensional Navier–Stokes equations are often scattered across the literature. Classical references on pseudospectral methods provide the mathematical foundations [16, 17], and many modern studies employ spectral DNS as a tool, yet there exists no concise, self-contained prescription that walks a new practitioner through the full workflow needed to reproduce a statistically stationary turbulent state: construction of the vorticity equation, choice of forcing, dealiasing, time stepping, initialization, diagnostic evaluation, and data handling.

The purpose of the present document is to fill this methodological gap. We provide a complete, reproducible, and fully documented pseudospectral DNS framework for simulating forced 2D turbulence, together with explicit formulas, algorithmic steps, implementation details, and diagnostics commonly used in the analysis of turbulent flows. To our knowledge, no comparable reference exists that simultaneously presents the physical model, numerical formulation, forcing construction, dealiased nonlinear evaluation, and diagnostic suite in a unified and pedagogically accessible manner. The material presented here is intended both as a practical guide for newcomers and as a transparent baseline implementation suitable for verification, benchmarking, and extensions to more advanced studies.

II. MODEL & ALGORITHM

We consider two–dimensional turbulence on a periodic square domain $\Omega = [0, L] \times [0, L]$. The grid contains N points in each direction, giving a uniform spacing $\Delta x = \frac{L}{N}$. The physical gridpoint coordinates are $X_{ij} = \frac{iL}{N}$, $Y_{ij} = \frac{jL}{N}; i, j = 0, \ldots, N-1$. All physical–space fields (vorticity, streamfunction, velocity, etc.) are defined on this grid. The solver stores the constants appearing in the 2D vorticity equation

$$\partial_t \omega + \mathbf{u} \cdot \nabla \omega = \nu \nabla^2 \omega - \mu \omega + f,$$

where ν is the viscosity, μ is the Ekman drag and dt is the timestep for temporal integration. The forcing is random and acts only on Fourier modes whose magnitude lies in a prescribed band $k_{\min} \leq |\mathbf{k}| \leq k_{\max}$. The forcing amplitude is normalized so that the real–space RMS forcing attains a desired value. All parameters are stored for constructing the forcing operator. Due to periodicity, admissible Fourier modes are indexed by $n_x, n_y \in \{-N/2 + 1, \ldots, N/2\}$. The corresponding wavenumbers are $k_x = \frac{2\pi}{L} n_x$, $k_y = \frac{2\pi}{L} n_y$, with magnitude and squared magnitude

$$|k| = \sqrt{k_x^2 + k_y^2}.$$

The streamfunction in Fourier space is obtained from vorticity via $\psi(\mathbf{k}) = -\frac{\omega(\mathbf{k})}{k^2}$ [16]. Since $k^2 = 0$ at the zero mode, the solver uses a modified quantity

$$k^2_{\text{safe}} = \begin{cases} k^2, & (n_x, n_y) \neq (0,0), \\ 1, & (n_x, n_y) = (0,0), \end{cases}$$

so that divisions never involve literal zeros. The solver analytically enforces $\psi(0,0) = 0$. Nonlinear terms such as $\mathbf{u} \cdot \nabla\omega$ generate unresolved modes that would alias back into the resolved spectrum. The spectral 2/3 rule applies the Galerkin projection

$$|n_x| \leq \frac{2}{3}\frac{N}{2}, \qquad |n_y| \leq \frac{2}{3}\frac{N}{2},$$

and sets all other modes to zero following each nonlinear product. This ensures mathematical consistency of the spectral method [17]. The solver initializes the vorticity field in Fourier space with $\omega(\mathbf{k}, 0) = 0$, which serves as the initial dynamical state to be advanced in time. The final call `init_forcing()` constructs the random forcing process in Fourier space.

A. Computation $\hat{\psi}$, $\hat{u}$ and $\hat{v}$

The code implements precisely the relation $\psi(\mathbf{k}) = -\frac{\omega(\mathbf{k})}{k^2}$, $\psi(0) = 0$. Velocity is recovered from the streamfunction via $u = \partial_y\psi$, $v = -\partial_x\psi$. Fourier differentiation gives $\widehat{\partial_y\psi} = ik_y\psi(\mathbf{k})$, $\widehat{\partial_x\psi} = ik_x\psi(\mathbf{k})$, so that the spectral velocity is

$$u(\mathbf{k}) = ik_y\psi(\mathbf{k}), \qquad v(\mathbf{k}) = -ik_x\psi(\mathbf{k}).$$

These expressions and guarantees divergence-free velocity $k_x u(\mathbf{k}) + k_y v(\mathbf{k}) = 0$ which gives an incompressible flow with $\nabla \cdot \mathbf{u} = 0$.

B. Scaled Forcing

The forcing is constructed in vorticity space as a frozen random spectrum supported on a prescribed wavenumber band. Define the forcing set $\mathcal{B} = \{\mathbf{k} : k_{\min} \leq |\mathbf{k}| \leq k_{\max}\}$. The solver implements this via a Boolean mask selecting only these modes. Each mode in $\mathcal{B}$ receives a complex number of unit magnitude,

$$\hat{g}(\mathbf{k}) = e^{i\theta(\mathbf{k})}, \qquad \theta(\mathbf{k}) \sim \text{Uniform}[0, 2\pi),$$

yielding an isotropic, phase-randomized forcing spectrum. Real physical-space forcing requires hermitian symmetry i.e., $\hat{f}_\omega(-\mathbf{k}) = \overline{\hat{f}_\omega(\mathbf{k})}$. The solver enforces this via reflection and averaging:

$$\hat{f}_\omega(\mathbf{k}) = \tfrac{1}{2}\big(\hat{g}(\mathbf{k}) + \overline{\hat{g}(-\mathbf{k})}\big),$$

ensuring that the inverse FFT yields a real field. The zero mode is suppressed $\hat{f}_\omega(0,0) = 0$, since forcing the mean vorticity would imply infinite circulation input. The forcing is finally projected onto the 2/3-dealiased region $\hat{f}_{\omega,\text{final}}(\mathbf{k}) = \hat{f}_\omega(\mathbf{k})\,\chi_{\text{dealias}}(\mathbf{k})$, eliminating all non-dealiased modes. Thus the frozen forcing spectrum stored by the solver is

$$\hat{f}_{\omega,\text{frozen}}(\mathbf{k}) = \begin{cases} e^{i\theta(\mathbf{k})}, & \mathbf{k} \in \mathcal{B} \text{ and dealiased}, \\ 0, & \text{otherwise}. \end{cases}$$

Beginning with $f_\omega(\mathbf{k}) = \hat{f}_{\omega,\text{frozen}}(\mathbf{k})$, the corresponding forcing streamfunction is computed using $\psi_f(\mathbf{k}) = -\frac{f_\omega(\mathbf{k})}{k^2}$. The physical-space forcing velocities follow from the streamfunction relations $f_x = \partial_y\psi_f$, $f_y = -\partial_x\psi_f$, which in Fourier space yield $\widehat{f_x} = ik_y\psi_f(\mathbf{k})$, $\widehat{f_y} = -ik_x\psi_f(\mathbf{k})$. Inverse transforms then give the real-space fields. The RMS magnitude of the forcing velocity is

$$\text{RMS} = \sqrt{\frac{1}{L^2}\int_\Omega (f_x^2 + f_y^2)\,dA} \approx \sqrt{\frac{1}{N^2}\sum_{i,j}(f_x^2 + f_y^2)}.$$

If this RMS value is nonzero, a scaling factor $\lambda = \frac{A_{\text{target}}}{\text{RMS}}$ is applied so that the resulting forcing satisfies $\sqrt{\langle(\lambda f_x)^2 + (\lambda f_y)^2\rangle} = A_{\text{target}}$ where A_{target} is the forcing amplitude tuned by user. Thus the final scaled vorticity-forcing spectrum returned each timestep is $\hat{f}^{\text{scaled}}_\omega(\mathbf{k}) = \lambda\,\hat{f}_{\omega,\text{frozen}}(\mathbf{k})$. The forcing is constructed as an isotropic, random, band-limited spectrum following the method of Eswaran and Pope [18].

C. Nonlinear Term

The nonlinear term in the 2D vorticity equation is

$$\mathcal{N}(\omega) = \mathbf{u} \cdot \nabla\omega = u\,\partial_x\omega + v\,\partial_y\omega.$$

The program computes this term in physical space and returns its Fourier transform with proper de-aliasing. It starts with recovering the stream function and velocity. The streamfunction in Fourier space is obtained from $\psi(\mathbf{k}) = -\frac{\omega(\mathbf{k})}{k^2}$, from which the velocity components follow $u(\mathbf{k}) = ik_y\psi(\mathbf{k})$, $v(\mathbf{k}) = -ik_x\psi(\mathbf{k})$. Using inverse FFTs $u(x,y) = \mathcal{F}^{-1}[u(\mathbf{k})]$, $v(x,y) = \mathcal{F}^{-1}[v(\mathbf{k})]$, and $\omega(x,y) = \mathcal{F}^{-1}[\omega(\mathbf{k})]$. Then we compute vorticity derivatives. Spectral differentiation gives $\partial_x\omega(\mathbf{k}) = ik_x\omega(\mathbf{k})$, $\partial_y\omega(\mathbf{k}) = ik_y\omega(\mathbf{k})$, and after inverse FFT,

$$\omega_x = \mathcal{F}^{-1}[ik_x\omega(\mathbf{k})], \;\; \omega_y = \mathcal{F}^{-1}[ik_y\omega(\mathbf{k})].$$

In physical space, $\mathcal{N}(x,y) = u\,\omega_x + v\,\omega_y$. In Fourier space $\widehat{\mathcal{N}}(\mathbf{k}) = \mathcal{F}[\mathcal{N}(x,y)]$. Using the 2/3-rule mask $D(\mathbf{k})$, $\widehat{\mathcal{N}}_{\text{dealias}}(\mathbf{k}) = D(\mathbf{k})\widehat{\mathcal{N}}(\mathbf{k})$. The Fourier-space vorticity equation is

$$\partial_t\omega(\mathbf{k}) = -\widehat{\mathbf{u} \cdot \nabla\omega} - \nu k^2\omega - \mu\omega + \hat{f}.$$

Thus the program returns $-\widehat{\mathcal{N}}_{\text{dealias}}(\mathbf{k})$, the correct nonlinear contribution to the RHS.

D. RK2 time stepping

We solve the Fourier-space PDE

$$\partial_t \omega(\mathbf{k}, t) = -\nu k^2 \omega - \mu\omega + N(\mathbf{k}, t) + f(\mathbf{k}, t),$$

where $N(\mathbf{k}) = -\widehat{\mathbf{u} \cdot \nabla\omega}$. Define the linear operator $L(\mathbf{k}) = \nu k^2 + \mu$, whose integrating factor is e^{Lt}. Now we precompute exponentials $E_{1/2} = e^{-L\Delta t/2}$, $E_1 = e^{-L\Delta t}$. Then perform

1. First Nonlinear Evolution : $N_1 = N(\omega) + f$.
2. Midpoint State : $\omega^\star = E_{1/2}\big(\omega + \frac{\Delta t}{2} N_1\big)$.
3. Second Nonlinear Evaluation : $N_2 = N(\omega^\star) + f$, with forcing re-evaluated if time-dependent or stochastic.
4. Final Update : $\omega^{n+1} = E_1\, \omega + \Delta t\, E_{1/2}\, N_2$.

This scheme is second-order accurate and integrates the stiff linear part exactly [19].

The program returns the physical fields

$$\big(\, \omega(x, y),\ u(x, y),\ v(x, y)\,\big).$$

These represent the real–space vorticity and velocity associated with the solver's current spectral state.

E. Initial Condition

The routine constructs an initial vorticity field $\omega(x, y, t = 0)$ whose Fourier coefficients are random but restricted to a narrow annulus in wavenumber space. This produces a smooth, band–limited random field commonly used as an initial condition in 2D turbulence simulations. If the user does not specify the central wavenumber k_0, it is taken from the forcing band $k_0 = \frac{1}{2}(k_{\min} + k_{\max})$, width $= \frac{1}{2}(k_{\max} - k_{\min})$. Otherwise the user–supplied k_0 is used and the width defaults to 1. The active spectral region is the annulus $\mathcal{B} = \big\{\mathbf{k} : \, ||\mathbf{k}| - k_0| \le \text{width}\big\}$. Let $k = |\mathbf{k}| = \sqrt{k_x^2 + k_y^2}$. Define

$$\chi_{\mathcal{B}}(\mathbf{k}) = \begin{cases} 1, & |k - k_0| \le \text{width}, \\ 0, & \text{otherwise}. \end{cases}$$

If no grid points satisfy this condition, the routine terminates with an error. The code generates complex Gaussian noise, $\eta(\mathbf{k}) = \eta_1(\mathbf{k}) + i\,\eta_2(\mathbf{k})$, $\eta_1, \eta_2 \sim \mathcal{N}(0, 1)$, and applies the spectral mask $\hat{\omega}(\mathbf{k}) = A\,\eta(\mathbf{k})\,\chi_{\mathcal{B}}(\mathbf{k})$. To obtain a real physical field, the coefficients must satisfy $\hat{\omega}(-\mathbf{k}) = \overline{\hat{\omega}(\mathbf{k})}$. The routine enforces this via the symmetrized assignment

$$\hat{\omega}(\mathbf{k}) \ \leftarrow\ \frac{1}{2}\Big(\hat{\omega}(\mathbf{k}) + \overline{\hat{\omega}(-\mathbf{k})}\Big),$$

and sets the zero mode to zero, $\hat{\omega}(0, 0) = 0$, so that the initial circulation vanishes. The current RMS velocity is

$$U_{\text{rms}}^{\text{current}} = \sqrt{\big\langle u^2 + v^2 \big\rangle}.$$

If the user specifies a target RMS velocity $U_{\text{rms}}^{\text{target}}$, the spectrum is rescaled by $\lambda = \frac{U_{\text{rms}}^{\text{target}}}{U_{\text{rms}}^{\text{current}}}$, $\hat{\omega}_{\text{final}}(\mathbf{k}) = \lambda\,\hat{\omega}(\mathbf{k})$. If the solver defines a de-aliasing mask $\chi_{\text{dealias}}(\mathbf{k})$, the initial condition is projected onto the admissible spectral region $\hat{\omega}_{\text{final}}(\mathbf{k}) \ \leftarrow\ \hat{\omega}_{\text{final}}(\mathbf{k})\,\chi_{\text{dealias}}(\mathbf{k})$. The program returns useful diagnostic information $U_{\text{rms}}^{\text{current}}, U_{\text{rms}}^{\text{target}}, \lambda, k_0, \text{width}, \#\{\mathbf{k} \in \mathcal{B}\}$. The mathematical form of the constructed vorticity field is

$$\omega_0(x, y) = \mathcal{F}^{-1}\Big[\lambda\, A\, \eta(\mathbf{k})\, \chi_{\mathcal{B}}(\mathbf{k})\, \chi_{\text{dealias}}(\mathbf{k})\Big].$$

This is a random, band–limited, real vorticity field suitable for initiating 2D turbulence simulations.

F. Parameters

The solver advances the two-dimensional vorticity equation $\partial_t\omega + \mathbf{u} \cdot \nabla\omega = \nu\nabla^2\omega - \mu\omega + f$ on a square periodic domain. The simulation parameters are $N = 256$, $L = 2\pi$, $\nu = 10^{-5}$, $\mu = 0.02$, and a timestep $\Delta t = 10^{-3}$. A statistically stationary flow is maintained using a random band-limited forcing of amplitude 0.03 applied at wavenumber $k_f = 5$. The total simulated time is $T = 1000$, corresponding to 10^6 timesteps. The initial condition is a band-limited random vorticity field centered at $k_0 = 2$ and scaled so that the initial RMS velocity satisfies $U_{\text{rms}} = 0.1$.

A function `compute_diagnostics` evaluates all standard quadratic measures used to quantify two-dimensional turbulence. The kinetic energy is $\text{KE} = \frac{1}{2}\langle u^2 + v^2\rangle$, while the enstrophy is $\mathcal{Z} = \frac{1}{2}\langle\omega^2\rangle$. Palinstrophy, defined as $P = \langle|\nabla\omega|^2\rangle$, is computed spectrally using $\omega_x = ik_x\omega(\mathbf{k})$ and $\omega_y = ik_y\omega(\mathbf{k})$. It provides the enstrophy-dissipation rate via $\epsilon_\omega = \nu P$. The forcing contributes an enstrophy injection rate $\epsilon = \langle\omega f\rangle$, where f is obtained by inverse transforming the spectral forcing field. These quantities are standard invariants used in the analysis of 2D turbulence [1, 3].

Several diagnostic Reynolds numbers are computed. The RMS velocity is $U_{\text{rms}} = \sqrt{\langle u^2 + v^2\rangle}$. The large-scale or "box" Reynolds number is $Re_{\text{box}} = U_{\text{rms}}L/\nu$, while the forcing-scale Reynolds number is $Re_f = U_{\text{rms}}/(\nu k_f)$, based on the forcing length scale $\ell_f = 1/k_f$. Velocity gradients obtained spectrally yield $\langle|\nabla u|^2\rangle = \langle u_x^2 + u_y^2 + v_x^2 + v_y^2\rangle$, from which the Taylor microscale $\lambda = U_{\text{rms}}/\sqrt{\langle|\nabla u|^2\rangle}$ and Taylor Reynolds number $Re_\lambda = U_{\text{rms}}\lambda/\nu$ are determined. These diagnostic quantities follow standard turbulence theory [4].

G. Time-Loop Outputs

Every 200 steps the solver performs the following tasks saves the full simulation state $(\omega, u, v, \omega_{\mathbf{k}}, t)$, prints all diagnostic quantities, prints all Reynolds numbers, monitors the evolution of enstrophy, energy, palinstrophy, and

the enstrophy injection rate. This produces a complete time-evolution dataset suitable for spectral analysis, cascade studies, and steady–state detection. The solver integrates the vorticity equation

$$\partial_t \omega + \mathbf{u} \cdot \nabla \omega = \nu \Delta \omega - \mu \omega + f.$$

The numerical scheme employs a spectral discretization, integrating-factor RK2 time stepping, random band-limited forcing, dealiasing using the 2/3 rule. At regular intervals the solver computes

$$\mathrm{KE} = \tfrac{1}{2}\langle u^2 + v^2 \rangle, \quad \mathcal{Z} = \tfrac{1}{2}\langle \omega^2 \rangle,$$
$$P = \langle |\nabla \omega|^2 \rangle, \quad \epsilon = \langle \omega f \rangle,$$
$$Re_{\mathrm{box}} = \frac{U_{\mathrm{rms}} L}{\nu}, \quad Re_f = \frac{U_{\mathrm{rms}}}{\nu k_f}, \quad Re_\lambda = \frac{U_{\mathrm{rms}}^2}{\nu \sqrt{\langle |\nabla u|^2 \rangle}}.$$

The full flow state is written to file for later analysis.

III. RESULTS

We now present the statistical and structural diagnostics obtained from the direct numerical simulation of forced two–dimensional turbulence. The results are interpreted in light of classical theories of the inverse–energy and forward–enstrophy cascades [1–3] and the modern understanding of intermittency, strain–vorticity dynamics, and velocity–gradient topology [12–15, 20].

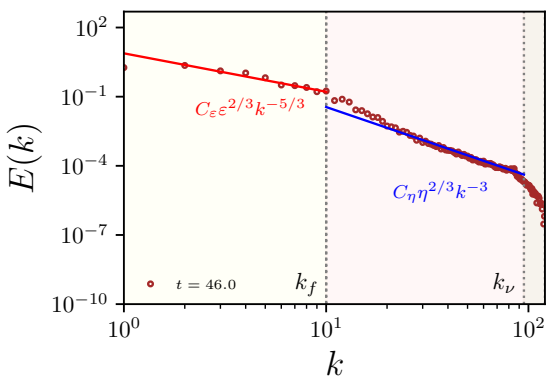

FIG. 1: Energy spectrum $E(k)$ from the two–dimensional Navier–Stokes simulation at $t = 46.0$. The forcing scale $k_f \approx 10$ (grey band) separates the large–scale inverse energy cascade, where the classical $k^{-5/3}$ law [1] is expected, from the forward enstrophy cascade that approaches a k^{-3} scaling in the sense of Batchelor [2]. For $k < k_f$ the measured spectrum shows a clear $k^{-5/3}$ behaviour, in agreement with contemporary reviews [3]. For $k > k_f$ the spectrum steepens toward k^{-3} before viscous damping becomes dominant.

IV. DISCUSSION

The results presented above demonstrate that the pseudospectral framework developed in this work successfully

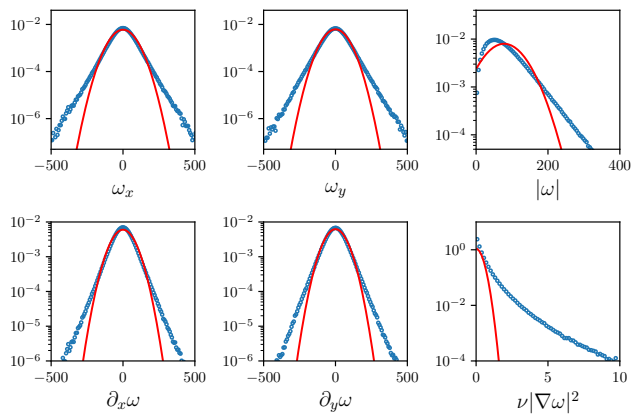

FIG. 2: Probability density functions of vorticity–related quantities computed from the final $N = 50$ snapshots. Heavy non–Gaussian tails in ω, its gradients, and the dissipation field $\nu|\nabla\omega|^2$ are consistent with the strong intermittency observed previously in 2D turbulence [14, 15]. The one–sided tail of the dissipation field reflects the localized, sheet–like vorticity–gradient structures typical of the forward enstrophy cascade.

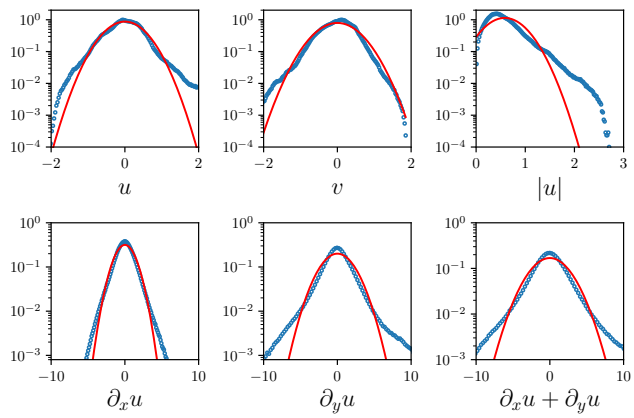

FIG. 3: Probability density functions of the velocity components and velocity–gradient fields, averaged over the final $N = 50$ snapshots. The pronounced non-Gaussian behaviour and stretched–exponential tails in both u, v and their gradients reflect the enhancement of intermittency by coherent vortices and shear layers, in agreement with prior observations [14, 15]. The gradient PDFs in particular deviate strongly from Gaussianity, indicating intense small–scale straining motions.

reproduces the principal statistical and structural features of forced two–dimensional turbulence. The canonical dual–cascade phenomenology—an upscale transfer of kinetic energy and a downscale transfer of enstrophy—is clearly reflected in the energy spectrum, which exhibits the expected $k^{-5/3}$ scaling range for $k < k_f$ and approaches the Batchelor k^{-3} prediction in the forward–cascade regime for $k > k_f$. The agreement with these classical theoretical scalings reinforces the accu-

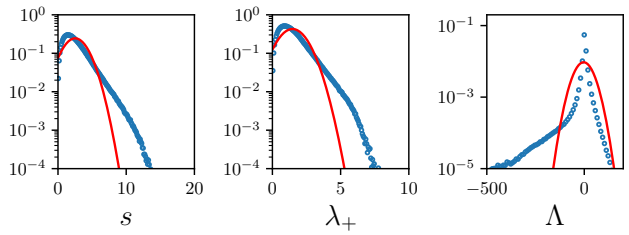

FIG. 4: Probability density functions of the strain magnitude s, the positive strain eigenvalue λ_+, and the Okubo–Weiss invariant $\Lambda = s^2 - \omega^2$. The skewed structure of the Λ distribution reproduces the well-known separation between strain–dominated and vorticity–dominated regions first emphasized by Okubo [12] and Weiss [13]. The heavy tails in s and λ_+ reflect the strong local deformation produced by shear events in the enstrophy cascade range.

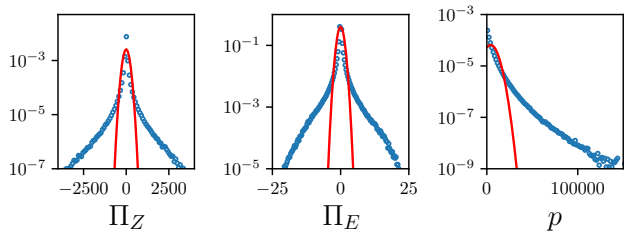

FIG. 5: Probability density functions of the nonlinear enstrophy flux Π_Z, the local energy flux Π_E, and the palinstrophy density $p = \frac{1}{2}|\nabla\omega|^2$. The extreme non-Gaussianity of Π_Z and p is characteristic of the intermittent, localized nature of vortex stretching and vorticity–gradient production in 2D turbulence [3, 14, 15]. The broad tails of Π_E highlight the strong fluctuations in the instantaneous energy transfer.

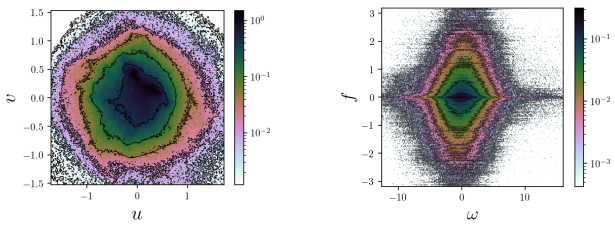

FIG. 6: Left: joint PDF $P(u, v)$ of velocity components, showing nearly isotropic but strongly non-Gaussian statistics, a hallmark of 2D turbulence with coherent vortices [15]. Right: joint PDF $P(\omega, f)$ of vorticity and forcing. The diamond–shaped structure reveals the preferred alignment between forcing and vorticity at injection scales [21], and rapid decorrelation at smaller scales.

racy of the dealiased nonlinear evaluation, the forcing construction, and the integrating–factor RK2 scheme implemented here.

The probability–density functions of vorticity, velocity, gradients, and dissipation fields further indicate that the simulation achieves a statistically stationary state

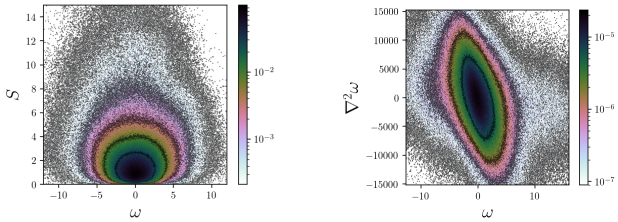

FIG. 7: Left: joint PDF $P(\omega, S)$ highlighting the strong statistical coupling between rotation and strain [14, 15]. The heavy tails reflect sporadic strain bursts and vortex–merger events. Right: joint PDF $P(\omega, \nabla^2\omega)$ reveals an almost elliptical structure, consistent with the localized, coherent–vortex–dominated dissipative dynamics described in classical 2D turbulence theory [1, 3].

marked by strong intermittency. Non–Gaussian tails, asymmetric strain statistics, and the heavy–tailed distributions of enstrophy–flux and palinstrophy density are consistent with earlier numerical and experimental studies [3, 14, 15]. These results confirm that the numerical framework is capable of resolving the intense, localized structures characteristic of the forward–enstrophy cascade, including thin shear layers and vortical filaments.

The joint probability distributions provide a complementary view of the organisation of the flow. The velocity JPDF shows a nearly isotropic but markedly non-Gaussian structure, while the vorticity–forcing and vorticity–strain JPDFs capture the dynamical coupling between coherent vortices, strain-dominated regions, and the forcing mechanism. Diagrams based on the Okubo–Weiss parameter and velocity–gradient invariants reveal the correct geometric partitioning of the flow into elliptic (vortical) and hyperbolic (strain-dominated) regions, reproducing the characteristic signatures established by Okubo, Weiss, and subsequent authors. Collectively, these diagnostics illustrate that the solver accurately captures not only the spectral dynamics but also the fine-scale topology and intermittency that define two–dimensional turbulence.

Perhaps more importantly, the results demonstrate that all of these physical signatures emerge from a compact and fully documented numerical implementation. Each step of the simulation—initialization, forcing construction, dealiased nonlinear evaluation, time stepping, and diagnostic computation—is explicitly specified, thereby enabling full reproducibility. This stands in contrast to many studies in the literature where the methodological details are implicit, distributed across multiple sources, or tailored to specialised research codes. The present work consolidates these components into a unified framework accessible to new practitioners while remaining robust enough to serve as a reference baseline for more advanced investigations.

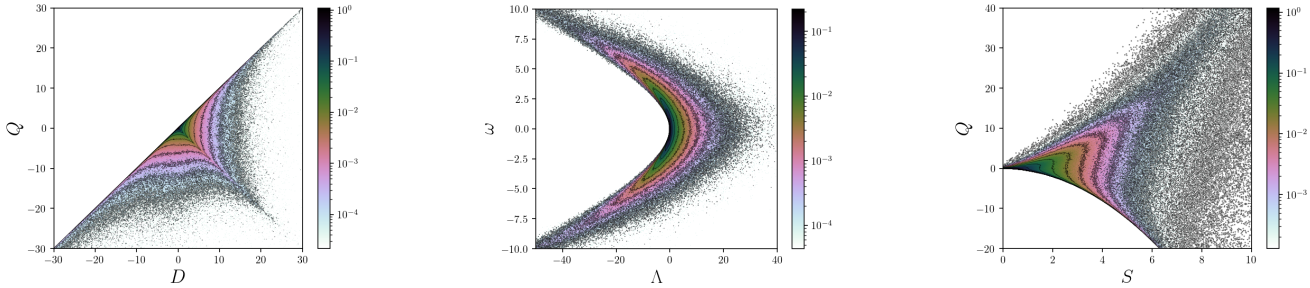

FIG. 8: Left: (D, Q) diagram representing the 2D analogue of the 3D Q–R analysis of velocity–gradient topology [20]. The triangular support reflects incompressibility constraints. Middle: the Okubo–Weiss diagram reproduces the classical strain–rotation classification of Okubo [12] and Weiss [13]. Right: (S, Q) diagram illustrating the algebraic constraint $Q = \frac{1}{2}(\omega^2 - S^2)$ and the strong asymmetry associated with coherent vortices.

V. CONCLUSION

This document provides a complete, reproducible, and self-contained prescription for performing direct numerical simulations of forced two–dimensional turbulence using a Fourier pseudospectral method. By assembling the essential components—formulation of the vorticity equation, construction and scaling of the forcing, dealiasing strategy, nonlinear term evaluation, integrating-factor time stepping, and comprehensive diagnostic tools—the work fills a methodological gap that has persisted despite the extensive literature on 2D turbulence.

The framework reproduces well-established features of 2D turbulent flows, including dual-cascade behaviour, coherent vortical structures, strong intermittency, and the geometric organisation of velocity–gradient dynamics. Its transparent formulation and modular design make it suitable for pedagogical use, code verification, and extension to more complex systems such as quasi-geostrophic turbulence, magnetohydrodynamic turbulence, or flows with active scalar coupling.

As a baseline implementation, the solver can serve as a starting point for systematic studies of spectral energy transfer, intermittency, anisotropy, coherent structure formation, and parameter dependence. Because every step of the algorithm is explicitly documented and easily reproducible, the framework also provides a reliable tool for benchmarking alternative numerical schemes or validating analytical results.

In summary, the present work aims to make the simulation of two–dimensional turbulence more accessible, transparent, and methodologically coherent. By providing a full end-to-end reference implementation, it enables researchers and students alike to explore, reproduce, and extend one of the most fundamental systems in fluid dynamics with clarity and confidence.

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